

Simulating Varied Indoor Lighting for Robust Real-vs-Fake Face Classification

Executive summary

A real-vs-fake face classifier that is stable under codec/platform effects but flips decisions with time-of-day lighting is exhibiting a classic *photometric domain shift*: the underlying identity content is similar, while illumination, white balance, exposure control, shadows, and specularities change the input distribution in ways your backbone was not optimised to be invariant to. This is a known failure mode for vision systems deployed “in the wild”, and deepfake/forgery detectors are also documented to degrade under real-world perturbations and distribution shifts. ¹

The augmentation space that matters for *indoor video conferencing / office / home* lighting is broader than “brightness/contrast jitter”: it includes (a) camera auto-exposure (AE) and auto white balance (AWB) dynamics, (b) correlated colour temperature (CCT) shifts and mixed illuminants, (c) directional lighting and cast shadows (key light from a window; overhead downlight; monitor light), (d) backlight/rim lighting, (e) flicker/rolling-shutter interactions common under AC-powered LEDs/fluorescents, and (f) camera ISP non-linearities (gamma/tonemapping), possibly compounded by platform “portrait light / adjust brightness” features. ²

From the literature and available tools, the most pragmatic, high-ROI path is a **tiered augmentation strategy**:

1. **Fast photometric + shadow augmentation** (GPU/on-the-fly), calibrated to indoor ranges and your cameras’ colour response. This is inexpensive and often yields large robustness gains. ³
2. **Face-aware directional lighting augmentation** using **low-order spherical harmonics (SH)** plus estimated normals/albedo from a pre-trained inverse-rendering network (e.g., SfSNet / 3D face reconstruction). This produces more realistic shading changes than global jitter while staying lighter than full neural rendering. ⁴
3. **Occasional high-fidelity relighting** (offline or low-probability online) using portrait relighting models or SDKs. These can emulate “video-conference portrait lighting” and challenging cast-shadow situations, but you must manage the risk of introducing synthetic artefacts that resemble manipulation cues. ⁵

For training, combine these with **consistency regularisation** (AugMix-style) and a **condition-stratified evaluation protocol** (per lighting regime, per camera, per platform filter) with statistical comparison (e.g., DeLong test for AUC) and calibration metrics (ECE/Brier). ⁶

Why indoor lighting breaks your classifier

Indoor lighting changes many signals that deep networks (including CLIP-style backbones) can inadvertently exploit as shortcuts: global intensity distribution, local contrast around facial micro-texture,

colour casts (CCT), shadow boundaries, highlight patterns on skin, and even noise statistics due to AE raising gain in low light. Many webcam and phone pipelines also apply dynamic AE/AWB with measurable time courses, so the same person can drift in colour and brightness over seconds when illumination changes. ⁷

In video-conference contexts, the platform and OS may further *normalise* or *beautify* lighting (e.g., “portrait light / adjust brightness”), which means your training distribution (lab/YouTube) can diverge from deployment even if the sensor is the same. ⁸

Finally, lighting interacts with *temporal artefacts*: many LEDs/fluorescents exhibit flicker at ~100/120 Hz (often twice mains frequency), which can alias into video depending on exposure timing and rolling shutter readout—creating banding or subtle brightness modulation across frames. ⁹

These factors are especially relevant for real-vs-fake classifiers because a portion of the evidence can live in subtle texture/frequency cues; aggressive or mis-calibrated relighting may either mask genuine forgery artefacts or create false “forgery-like” artefacts. Evidence that forgery detectors degrade under perturbations and distribution shift is well documented in deepfake benchmarks that introduce real-world distortions/perturbations. ¹⁰

Methods landscape from augmentation to relightable inverse rendering

Photometric augmentation and robustness-oriented training

Basic photometric transforms (brightness/contrast/saturation/hue/gamma, noise, and shadows) are standard building blocks in modern augmentation toolchains (torchvision, Albumentations, Kornia, NVIDIA DALI). ¹¹

Robustness-oriented schemes such as AutoAugment/RandAugment search or randomise augmentation policies, while AugMix explicitly combines random augmentations and enforces prediction consistency, improving robustness and uncertainty estimates on corruption benchmarks. ¹²

Domain-randomisation is the broader principle of training over sufficient synthetic variability so the real world appears as “just another variation”. ¹³

Face relighting, intrinsic decomposition, and inverse rendering

Face-specific relighting research spans:

- **End-to-end portrait relighting**: “Deep Single-Image Portrait Relighting” trains a CNN to relight a portrait given target SH lighting; it reports relighting up to 1024×1024 and explicitly uses spherical harmonics lighting representations. ¹⁴
- **Physics-guided face relighting**: CVPR 2020 “Learning Physics-guided Face Relighting under Directional Light” explicitly decomposes faces into intrinsic components under a diffuse physics-based image formation model and adds a learned correction for non-diffuse effects (specularities/shadows), trained/evaluated with light-stage captures. ¹⁵
- **Inverse rendering / decomposition for faces**: SfsNet decomposes an unconstrained face image into *surface normals*, *albedo*, and *lighting* using a physical Lambertian rendering model and a training

paradigm mixing synthetic labelled and real unlabelled data; it also uses SH lighting in its model.

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- **Video relighting and temporal consistency:** “Neural Video Portrait Relighting in Real-Time via Consistency Modeling” describes a large video relighting dataset with temporal OLAT images and environment lighting maps, and targets real-time coherent video relighting under dynamic illumination. 17
- **3D-aware / neural rendering approaches:** newer systems (NeRF / tri-plane / Gaussian head avatars) are increasingly real-time and controllable, and are explicitly motivated by applications like video conferencing. 18

A key practical bridge between “simple jitter” and “full neural rendering” is **low-order spherical harmonics lighting**: for approximately Lambertian surfaces, low-frequency lighting effects can be represented compactly (often using the first 9 SH components), and this underpins many face lighting models and relighting pipelines. 19

Practical techniques and recommended indoor lighting recipes

High-throughput photometric and shadow pipeline

Use a *face-crop-first* pipeline (detect/align, then augment the crop) to keep compute bounded and keep the background from becoming a confounder unless your deployment includes meaningful background cues. The following operations are widely available and fast:

- **Brightness/contrast/saturation/hue jitter** in torchvision / Albumentations / Kornia / DALI. 20
- **Gamma jitter** (captures ISP/gamma/tonemapping drift). 21
- **Synthetic shadows** (to mimic directional light occlusion). Albumentations exposes RandomShadow/PlasmaShadow-style transforms. 22
- **Sensor/noise transforms** (ISO-like noise) that correlate with low-light AE gain. 23

Colour temperature and mixed-illuminant augmentation

Indoor lighting spans warm (incandescent) to cool (daylight/overcast), and offices/homes frequently sit in the 2700K–6500K range with mixed sources (window daylight + overhead LEDs), which interacts with AWB assumptions. 24

A practical approach is to simulate **AWB drift** by applying *channel gains* in linear RGB corresponding to a sampled CCT (and optionally blending two CCTs across the image to emulate mixed illuminants). Even approximate Kelvin→RGB mappings can be adequate for augmentation if you clamp to plausible ranges and include sensor-specific calibration (below). 25

Directional lighting via spherical harmonics

To mimic “window from the left”, “ceiling downlight”, etc., generate SH coefficients representing a directional lobe + ambient term, render shading on an estimated normal map, and then composite with a face-specific albedo estimate. This is grounded in the classical observation that Lambertian reflectance under distant illumination is well approximated in a low-dimensional SH subspace, and it is explicitly used by face decomposition and portrait relighting systems. 26

Relighting via estimated normals/albedo or neural relighting models

You can increase realism by recovering normals/albedo from an inverse-rendering model and re-rendering under new lighting. Options include:

- **SfSNet-style decomposition**, yielding normals, albedo, SH lighting and enabling relighting. ²⁷
- **3D face reconstruction models** that explicitly estimate lighting parameters from a single image (e.g., DECA's codebase advertises predicting lighting and detailed geometry). ²⁸
- **Portrait relighting networks** (e.g., Deep Single-Image Portrait Relighting) that accept a target SH lighting vector and generate a relit portrait. ²⁹
- **Real-time video relighting** and 3D-aware methods (NeRF/tri-plane), which report real-time fps and explicitly aim at video-conference-like inputs. ³⁰
- **Commercial/SDK-grade relighting**: the NVIDIA Maxine “Video Relighting” filter takes an input frame, a matte, and an HDRI environment map, and outputs a relit foreground (intended for real-time pipelines). ³¹

Indoor lighting parameter ranges and “recipes” (starting point)

The table below is intentionally **actionable**: it gives ranges that (a) align with common library parameterisations and (b) cover plausible indoor variability. Tune them by matching your target cameras’ histograms/colour casts.

Phenomenon to mimic	Suggested augmentation (face crop unless noted)	Practical starting ranges (sample uniformly unless stated)	Notes / rationale
Low-light / overexposure drift	Brightness/contrast + gamma	brightness $\times$ [0.6, 1.4]; contrast $\times$ [0.7, 1.5]; gamma $\in$ [0.7, 1.6]	Matches common “jitter” knobs and gamma non-linearity; gamma correction is a standard modelling axis for display/camera pipelines. ³²
Warm vs cool white point	CCT→RGB gains (AWB)	CCT $\in$ [2700K, 6500K]; occasional extremes [2200K, 8000K] with low probability	Typical residential/office/daylight ranges; mixed-illuminant AWB is a known challenge for AWB systems. ²⁴
Mixed lighting (window + lamp)	Spatially varying colour cast	Blend two CCTs with a smooth mask; mix weight $\alpha \in$ [0.2, 0.8]	Emulates non-global illuminant; aligns with mixed-illuminant AWB framing. ³³
Directional key light & shadows	SH-based shading on normals, or RandomShadow	SH: ambient term + 1–2 directional lobes; shadow intensity 0.3–0.7	Low-order SH is standard for diffuse lighting; RandomShadow explicitly simulates regional darkening. ³⁴

Phenomenon to mimic	Suggested augmentation (face crop unless noted)	Practical starting ranges (sample uniformly unless stated)	Notes / rationale
Backlight / rim light	Add directional lobe from behind + highlight boost	Back-direction lobe prob 0.2; add mild specular proxy	Key in office “window behind” cases; realistic cast shadows/specularities are emphasised in modern relighting work. ³⁵
Platform “portrait light”	Mild face-only brighten + soften	Apply as <i>extra branch</i> for a subset of samples ($p \approx 0.2-0.4$)	OS/platforms advertise portrait lighting correction; emulate as a domain factor. ⁸
Flicker / banding	Temporal sinusoidal modulation + rolling-shutter stripes	Frequency 100/120 Hz; amplitude small (e.g., 1–5% luminance); apply to sequences	100/120 Hz flicker is common under AC lighting and documented; camera makers document anti-flicker targeting these bands. ⁹
Camera ISP differences	Per-camera colour matrix / tone curves	Learn per-camera affine in log-RGB; fit from target samples	AWB and ISP vary by device and are a major domain factor. ³⁶

Camera-specific synthesis (recommended): treat each deployment camera as its own “style”. If you can capture a small calibration set per camera (real + fake, multiple lighting states), fit simple per-camera transforms: (1) RGB channel gains distribution (AWB), (2) tone curve/gamma distribution, (3) noise vs brightness curve (shot/read noise proxy). This mirrors the reality that AE/AWB have device-dependent dynamics and that anti-spoofing benchmarks explicitly treat “sensor/device” as a covariate affecting generalisation. ³⁶

Tools, libraries, and off-the-shelf relighting models

Comparison table: integration-relevant properties

The goal here is not “best relighting paper”, but what is easiest to operationalise for augmentation.

Tool / model family	Primary input → output	Supervision / annotations required	Pretrained availability (practical)	Compute & speed (practical)	Integration notes for augmentation
torchvision / Albumentations / Kornia / DALI photometric ops	RGB → RGB	None	Yes (library) ²⁰	Very low; can be on-the-fly (GPU feasible) ³⁷	Best “first line”: easy, fast, controllable; pair with condition labels for evaluation.

Tool / model family	Primary input → output	Supervision / annotations required	Pretrained availability (practical)	Compute & speed (practical)	Integration notes for augmentation
Deep Single-Image Portrait Relighting (DPR, ICCV 2019)	portrait + target SH → relit portrait	Training uses generated dataset; at inference needs target SH	Code/data referenced in paper and repo exists ²⁹	CNN inference; paper reports 1024×1024 capability ³⁸	Good for <i>offline</i> generation of relit crops across a lighting grid; manage artefacts.
SfSNet (CVPR 2018) decomposition	face image → normals + albedo + lighting (+ reconstruction)	Mix of labelled synthetic + unlabelled real (method-specific) ²⁷	Research code linked in paper (historically) ³⁹	Medium; produces intrinsic maps	Use as intermediate: estimate normals/albedo then apply your own SH relighting on-the-fly.
Physics-guided directional face relighting (CVPR 2020)	face → intrinsic + directional relit	Requires light stage data for training/eval in paper ⁴⁰	Research project page; less “plug-and-play”	Higher; aims at realistic shadows/specularities ⁴⁰	Strong realism, but high integration cost; consider as offline augmentation if weights exist.
Real-time 3D-aware portrait video relighting (CVPR 2024 highlight)	portrait video → relit video (novel light/view)	Model-specific (NeRF-based, temporal consistency)	Official repo; reports runtime	Reported 32.98 fps on consumer hardware ⁴¹	Excellent for generating realistic “video-call style” relit sequences offline; verify artefact profile.
PN-Relighting (IJCV 2023, repo)	image (+ masks/env) → geometry/albedo → relit	Needs masks and env maps; provides checkpoints download instructions ⁴²	Checkpoints via linked storage	Medium; multi-stage pipeline	Useful if you can supply masks and target env maps; best as offline generator.

Tool / model family	Primary input → output	Supervision / annotations required	Pretrained availability (practical)	Compute & speed (practical)	Integration notes for augmentation
NVIDIA Maxine Video Relighting SDK	input RGB + matte + HDRI → relit foreground	Requires a matte (segmentation) and HDRI env map ⁴³	SDK + sample apps available ⁴⁴	Real-time on NVIDIA GPU; TensorRT-optimised pipeline ⁴⁵	Strong “industry-grade” option: can generate target-domain augmentations close to conferencing use cases.

Notes on “links” and primary sources

Most of the above have primary sources accessible via the citations (papers and official repositories). In particular, GitHub repos for real-time relighting and PN-Relighting provide concrete install/run instructions and checkpoint logistics. ⁴⁶

Commercial/SDK-grade relighting is explicitly documented as taking HDRI environment maps and mattes, which aligns well with synthesising “office/home lighting rigs” as HDRIs. ⁴⁷

(Also relevant to your motivating domain) video platforms and OS stacks advertise “portrait light / adjust brightness” features; you can treat these as *domain transforms* to emulate in augmentation. ⁸

Implementation details and integration with a CLIP-backbone classifier

What to augment, where, and how often

A robust operational pattern is:

- **Augment after face detection/alignment** (so transforms act on the face manifold, not on varying backgrounds), then feed into your CLIP backbone.
- Use **multi-view consistency**: for each original crop, generate 2–3 augmented views and enforce consistent prediction/embedding, inspired by AugMix’s “consistency across diverse augmentations”. ⁴⁸
- Prefer **on-the-fly** for cheap transforms; do **offline** for expensive relighting models (NeRF/diffusion/SDK). This reduces training latency variance and lets you curate artefact risk.

On-the-fly vs offline: a practical split

On-the-fly (every batch): brightness/contrast/saturation/hue/gamma, mild noise, mild shadow masks. Libraries support this directly, including video-capable transforms in torchvision v2 and GPU augmentation in Kornia/DALI. ⁴⁹

Offline (curated, lower rate): generate relit variants from (a) SH relighting on SfSNet/DECA normals, (b) portrait relighting networks, or (c) SDK relighting with HDRIs. This is where you can encode richer scenarios (window backlight, multi-source lighting, strong cast shadows). ⁵⁰

Augmentation mixture ratios (starting point)

Because your task is forensic (real vs fake), you want to avoid “washing out” manipulation evidence. A conservative starting schedule is:

- 60–70% samples: mild photometric jitter only
- 20–30% samples: photometric + shadow + CCT/AWB drift
- 5–10% samples: SH-directional relighting or high-fidelity relighting (offline pool)

This aligns with the idea that augmentation strength is a tunable regulariser (RandAugment explicitly parameterises augmentation strength with a small search space). ⁵¹

Adversarial / worst-case augmentation

If lighting is the dominant deployment shift, you can *target it* by using adversarial composition of random augmentations to find perturbations that maximally degrade training loss while remaining within a defined augmentation set. This complements “random coverage” and can accelerate robustness. ⁵²

Pseudocode for a face-lighting augmentation pipeline

```
# Pseudocode sketch (mix of on-the-fly & offline ideas)

def augment_face_crop(img_rgb_uint8, rng):
    # 1) Convert to float [0,1] and optionally linearise (approx)
    x = to_float01(img_rgb_uint8)

    # 2) Cheap photometric jitter (always available in torchvision/
    albumentations/kornia)
    x = random_brightness_contrast(x, brightness=(0.6, 1.4), contrast=(0.7,
    1.5), p=0.8)
    x = random_gamma(x, gamma=(0.7, 1.6), p=0.5)
    x = random_hsv_jitter(x, hue_delta=(-0.05, 0.05), sat_mult=(0.7, 1.3),
    p=0.5)

    # 3) Colour temperature / AWB drift (custom)
    if rng.rand() < 0.6:
        cct1 = rng.uniform(2700, 6500)
        gains1 = kelvin_to_rgb_gains(cct1) # apply in approx-linear RGB
        x = apply_channel_gains(x, gains1)

    # optional mixed illuminant
    if rng.rand() < 0.3:
        cct2 = rng.uniform(2700, 6500)
```



```

        gains2 = kelvin_to_rgb_gains(cct2)
        mask = smooth_random_gradient_mask(x.shape[:2], rng) # 0..1
        x = x * (mask * gains1 + (1 - mask) * gains2)

# 4) Shadows / directional shading
if rng.rand() < 0.4:
    x = random_shadow_mask(x, intensity=(0.3, 0.7))

# 5) Optional SH relighting (requires normals+albedo, e.g., from SfSNet/DECA
offline)
    # if rng.rand() < 0.1:
    #     n, albedo = pretrained_inverse_render(x) # normals, albedo
    #     sh = sample_indoor_sh_lighting(rng)      # ambient + 1-2 lobes
    #     shading = render_lambertian_sh(n, sh)
    #     x = clamp01(albedo * shading + residual_specular_proxy(x, shading))

return to_uint8(x)

```

The “cheap jitter” functions correspond to widely used primitives in torchvision/Albumentations/Kornia, RandomGamma and RandomShadow are explicitly supported in Albumentations, and SH lighting is a standard representation for face illumination in face decomposition/relighting literature. ⁵³

Mermaid: end-to-end training pipeline for lighting robustness

```

flowchart TD
    A[Video frames] --> B[Face detect + align + crop]
    B --> C{Augmentation router}
    C -->|p~0.7| D[Fast photometric jitter\nbrightness/contrast/gamma/HSV + mild noise]
    C -->|p~0.2| E[Photometric + shadow + CCT/AWB drift\n(mixed illuminants)]
    C -->|p~0.1 offline pool| F[Relighting augmentation\nSH relight or neural/SDK relight]
    D --> G[Normalise + resize]
    E --> G
    F --> G
    G --> H[CLIP-style backbone]
    H --> I[Classifier head]
    I --> J[Loss: BCE/CE]
    H --> K[Optional consistency loss\n(2-3 augmented views per sample)]
    J --> L[Optimiser]
    K --> L

```

Evaluation protocol and experimental plan

Datasets and splits to reflect deployment lighting

Because the failure is *time-of-day / indoor lighting* on the same device, you should evaluate with **condition-labelled splits** that prevent leakage:

- **Per-subject split** (never mix identities across train/test if the identity distribution matters).
- **Per-session / per-lighting split**: train on some lighting regimes; test on held-out regimes. The *OULU-NPU* PAD benchmark is an instructive precedent: it explicitly treats illumination/background, sensors, and attack instruments as covariates, and defines protocols that introduce previously unseen conditions at test time. ⁵⁴
- **Per-camera split** (leave-one-camera-out) if you deploy across devices; again, *OULU-NPU*'s LOCO protocol is a relevant template. ⁵⁵

For deepfake/forgery detection datasets, include at least one benchmark that already incorporates real-world perturbations (e.g., *DeeperForensics-1.0* applies “real-world perturbations” and explicitly includes mixtures like blur + JPEG + noise + saturation changes). ⁵⁶

Metrics

Report metrics at three levels:

- **Discrimination**: AUC ROC, and optionally EER (common in biometric-style evaluation). ⁵⁷
- **Calibration / confidence**: ECE (expected calibration error) and Brier score loss. ⁵⁸
- **Stratified robustness**: compute all metrics **per lighting bin** (e.g., warm/cool; low/high exposure; backlit vs front-lit) and **per camera/platform condition**.

To compare AUCs between models on the same test set, use a correlated ROC comparison such as DeLong's nonparametric test. ⁵⁹

Ablations and sample sizing (practical plan)

Because your goal is to isolate lighting robustness, ablations should:

1. Baseline (existing codec/platform augmentation only)
2. ◦ photometric jitter (mild)
3. ◦ calibrated CCT/AWB drift
4. ◦ shadows
5. ◦ SH directional shading (normals-based)
6. ◦ high-fidelity relighting (offline pool)
7. ◦ consistency regularisation (AugMix-style) ⁴⁸

Use a fixed compute budget per run; RandAugment is useful as a “single-knob” way to tune augmentation strength when you want a systematic sweep rather than hand-tuning. ⁵¹

Mermaid: experiment timeline

```
gantt
    title Lighting Robustness Experiments
    dateFormat YYYY-MM-DD
    axisFormat %d %b

    section Data & instrumentation
    Capture target-domain eval clips (label lighting bins) :a1, 2026-03-10, 14d
    Build condition metadata (CCT proxy, exposure proxy, backlight flag) :a2,
    after a1, 7d

    section Baselines
    Reproduce current model + evaluation harness :b1, 2026-03-10, 7d
    Add per-condition reporting + calibration metrics :b2, after b1, 7d

    section Augmentation iterations
    Photometric + gamma + noise sweep :c1, 2026-03-24, 10d
    Add CCT/AWB drift + mixed illuminant :c2, after c1, 10d
    Add shadow + directional cues (SH or masks) :c3, after c2, 10d

    section Higher-fidelity relighting
    Offline relighting pool generation (choose method/SDK) :d1, 2026-04-23, 14d
    Integrate + rerun ablations :d2, after d1, 10d

    section Analysis
    Statistical tests (DeLong/bootstraps) + failure review :e1, 2026-05-17, 10d
    Final recommendation + training recipe lock-in :e2, after e1, 7d
```

Failure modes, biases, and ethical considerations

Technical failure modes

1. **Augmentation-induced forensic artefacts:** strong relighting networks (or aggressive tone manipulation) can create boundaries, halos, or texture shifts that resemble manipulation artefacts, risking false positives/negatives. This risk is real in systems trained to detect subtle inconsistencies, and deepfake detectors are known to be sensitive to perturbations and processing operations. ⁶⁰
2. **Shortcut re-learning:** if “dark = fake” becomes predictive in your augmented training distribution (e.g., if your fakes are over-represented in low light), the model may become *more* biased. Condition-balanced augmentation rates per class are important. The need for protocols that test unseen covariates is a central theme in anti-spoofing benchmarks. ⁵⁵
3. **Temporal inconsistency:** frame-wise augmentation that varies independently per frame can introduce unrealistic flicker, which may teach the model the wrong invariances; if you add flicker augmentation, do it as a coherent temporal process. Flicker/rolling shutter is a real phenomenon with known frequencies (100/120 Hz regimes). ⁶¹

Bias and fairness

Lighting interacts with skin reflectance and camera AWB behaviour, and AWB itself is an active research topic (including handling mixed illuminants and portrait-specific considerations). ⁶²

Face PAD literature also highlights demographic and domain biases (e.g., cross-ethnicity evaluation protocols), underscoring that robustness work should include subgroup analysis if your deployment has diverse users. ⁶³

Privacy and consent

If you collect office/home webcam video for evaluation or fine-tuning, treat it as sensitive biometric data. A good precedent is datasets collected with consented participants (e.g., DeeperForensics describes paid/consented actors in its dataset documentation). ⁶⁴

Minimise retention, store only derived face crops if possible, and ensure clear user consent and purpose limitation.

Clarifying questions to answer next

What will most improve the precision of the recommendations is knowing your target domain and constraints. Please answer:

1. What is the *deployment setting*: captured from a dedicated webcam, laptop camera, phone front camera, or a mixture? Are you deploying across multiple camera models? (Because device/AWB/ISP variability can dominate.) ⁶⁵
2. Do you control the *preprocessing*: can you lock exposure/white balance at capture time, or are you stuck with dynamic AE/AWB? (This determines whether to model temporal drift.) ⁶⁶
3. Is the classifier purely frame-based, or do you aggregate temporally (e.g., average logits, temporal model)? (This affects flicker/shadow temporal augmentation strategy.) ⁶⁷
4. What are the *error costs*: is false positive (“real flagged fake”) worse than false negative? Do you need calibrated probabilities? (Determines emphasis on calibration metrics and thresholding.) ⁶⁸
5. Do you have any *target-domain labelled data* (even small), and if so roughly how many subjects and lighting regimes? (Determines whether to add domain-adaptive fine-tuning beyond augmentation.) ⁶⁹
6. Are your “fakes” mostly deepfakes (GAN/diffusion face swaps) or presentation attacks (replays/prints), or both? (Lighting interacts differently with replay glare vs synthetic artifacts.) ⁷⁰
7. Do you run any platform-level video enhancement (portrait light/brightness filters), e.g., via Microsoft Teams ⁷¹ ? If yes, can you record both pre- and post-filter streams? ⁷²

¹ ¹⁰ ⁵⁶ [openaccess.thecvf.com](https://openaccess.thecvf.com/content_CVPR_2020/papers/Jiang_DeeperForensics-1.0_A_Large-Scale_Dataset_for_Real-World_Face_Forgery_Detection_CVPR_2020_paper.pdf)

https://openaccess.thecvf.com/content_CVPR_2020/papers/Jiang_DeeperForensics-1.0_A_Large-Scale_Dataset_for_Real-World_Face_Forgery_Detection_CVPR_2020_paper.pdf

² ⁷ ³⁶ ⁶⁶ [Automatic exposure and white balance control in video ...](https://www.researchgate.net/publication/260742083_Automatic_exposure_and_white_balance_control_in_video_cameras_Time_course_characterization_and_preference?utm_source=chatgpt.com)

[https://www.researchgate.net/publication/](https://www.researchgate.net/publication/260742083_Automatic_exposure_and_white_balance_control_in_video_cameras_Time_course_characterization_and_preference?utm_source=chatgpt.com)

[260742083_Automatic_exposure_and_white_balance_control_in_video_cameras_Time_course_characterization_and_preference?utm_source=chatgpt.com](https://www.researchgate.net/publication/260742083_Automatic_exposure_and_white_balance_control_in_video_cameras_Time_course_characterization_and_preference?utm_source=chatgpt.com)

3 11 20 32 53 **ColorJitter — Torchvision main documentation**

https://docs.pytorch.org/vision/main/generated/torchvision.transforms.ColorJitter.html?utm_source=chatgpt.com

4 16 27 39 50 **openaccess.thecvf.com**

https://openaccess.thecvf.com/content_cvpr_2018/papers/Sengupta_SfSNet_Learning_Shape_CVPR_2018_paper.pdf

5 14 29 38 **openaccess.thecvf.com**

https://openaccess.thecvf.com/content_ICCV_2019/papers/Zhou_Deep_Single-Image_Portrait_Relighting_ICCV_2019_paper.pdf

6 48 **AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty**

<https://openreview.net/pdf/bf92dcddbadea1e595d505ab2eaf501463464237.pdf>

8 **Windows Studio Effects Overview - Windows apps**

https://learn.microsoft.com/en-us/windows/apps/develop/windows-integration/studio-effects?utm_source=chatgpt.com

9 61 67 **Light Flicker from LED Lighting Systems**

https://www.led-professional.com/media/resources-1_articles_lighting-flicker-from-led-lighting-systems_LpR53_p50-p59.pdf/%40%40download/file/LpR53_p50-p59.pdf?utm_source=chatgpt.com

12 **AutoAugment: Learning Augmentation Strategies From Data**

https://openaccess.thecvf.com/content_CVPR_2019/papers/Cubuk_AutoAugment_Learning_Augmentation_Strategies_From_Data_CVPR_2019_paper.pdf

13 **Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World**

https://arxiv.org/abs/1703.06907?utm_source=chatgpt.com

15 40 **lvsn.github.io**

https://lvsn.github.io/face-relighting/2020CVPR_Nestmeyer-et-al_face-relighting.pdf

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